

3. Detailed report on the status of DTs selected for Capacitor scale-up

3.1 Methodology for Capacitor Usage Demonstration

Capacitor usage demonstration is a participatory exercise which needs the active involvement of all stakeholders, that is farmers, MSEDCL officials and agriculture officials. Awareness building is also an integral part of the Capacitor demonstration exercise so that it reaches a wider agricultural population. The following procedure is being followed for a capacitor demonstration in 10 different DTs in two districts.

3.1.1 Village Overview Meetings

Different villages from the Beed and Wardha districts were identified based on secondary data. Identified villages were visited to understand irrigation electricity status. Focused group discussions were conducted in villages to understand electricity problems. Those villages which were confirmed to have irrigation electricity problems due to overloading were finalized for capacitor demonstration.

3.1.2 Training of MSEDCL and Agri officials

Workshop at IIT Bombay was organized to conduct training of MSEDCL and Agriculture officials. The purpose of creating a knowledgeable person at the village level. As a part of this workshop, besides MSEDCL and Agri department officials, different stakeholders including linemen, zoro linemen, Village Sarpanch, and farmers, were given detailed information on capacitor usage. They were trained to install capacitors on irrigation pumps.



Figure 3.1 Workshop conducted in IIT Bombay

3.1.3 Material for awareness building and capacitor scale up

The purpose of selecting a limited number of DTs for capacitor scale-up was to create successful cases and make farmers aware of such cases. Farmers on selected DTs were provided with all necessary support for capacitor installation and awareness. Focused group discussions were conducted with farmers of selected DTs. FGDs were made aware of the function of the capacitor, its usage and benefits.



Figure 3.2 IITB team member explaining to villagers the importance of capacitors and load management to villagers



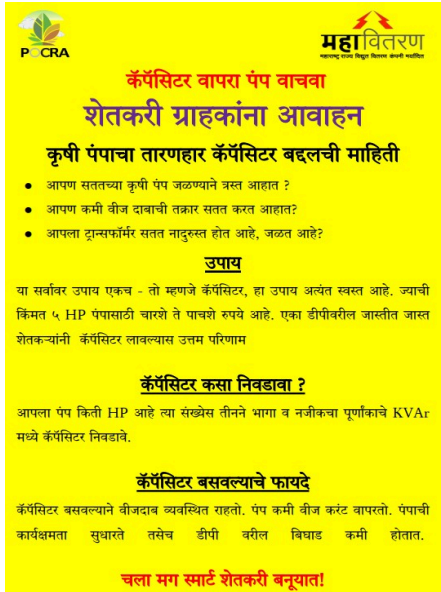
Figure 3.3 IITB team field visit to the DTs of Bagpimpalgaon

Capacitor awareness material that was made and circulated through different channels. All important stakeholders were communicated and farmers on selected DTs as well farmers from nearby villages made aware regarding capacitors.

Table 3.1 Stakeholders communication and awareness building reach

	To whom they can reach	People reached by IIT energy team
MSEDCL	- All consumers (Only text message) - EE can reach all officials from district - Officials WhatsApp group - Union of Lineman - WhatsApp group	- Feeder level farmer groups (200 farmers from each district) - Union of Lineman (1000 lineman)
Agri Department	- Krishi sahayak - WhatsApp groups of farmers (appr. 1000 farmers) - Agri Officials list	Around 5000 Farmers from each district
Sarpanch	Sarpanch - WhatsApp group of villagers (almost all farmers)	Almost all farmers of selected villages

Following awareness-building material was circulated among farmers:

	
Figure 3.4 Banner put in different locations (GP office, MSEDCL premises) to spread the word regarding capacitor demonstration	Figure 3.5 Capacitor awareness messages spread through smartphones by JEs/AEs (to the whatsapp groups of Zero Wiremen, Linemen and JE's along with few selected villagers)

3.1.4 Capacitor Availability

The structured demonstration of the value of capacitors is one of objectives, stated in MOU V, so it was decided that capacitors had to be given to farmers. Hence capacitors were purchased in a bulk. Going forward, if farmers are to buy capacitors, or a drive is to be conducted, then this needs to be kept in mind. Even though farmers are aware of the use of capacitors they need to be accessible to farmers easily. Presently the three phase capacitors (as opposed to the capacitors used by farmers for single phase or two-phase to three phase conversion) are available in the local market in limited places or in small numbers. So it was necessary to make them available to farmers on the selected DTs. Based on primary data collected about all pumps on selected DTs, capacitors with different ratings were ordered for farmers.

Firstly, a capacitor demonstration was carried out on a few pumps. Farmers were asked to attend demonstrations. During the demonstration, farmers were taught to install capacitors on their pumps. Capacitors were directly given to farmers suitable to their pump ratings and asked them to install by themselves. Farmers whatsapp groups were formed and guidance was provided for installation. Linemen and Zero linemen were asked to install capacitor of those farmers who could not install by themselves. At few places IIT Bombay energy team installed installed capacitors on farm pumps. IIT Bombay energy team kept track of capacitor installation on selected DTs.



Figure 3.6 Photo showing capacitor distribution to farmers

3.1.5 Capacitor demonstration with on-field experiments

Capacitor usage on agriculture pumps results in voltage improvement and reduction in current. The benefits of capacitor usage on irrigation were demonstrated to farmers by taking readings of drop in current and increase in voltage on agri pumps before and after capacitor installation. Several readings were taken on selected DTs. Table 3.2 shows an example of one such demonstration by measurements taken on Bagh DT, Tigaon.

Table 3.2 Measurements taken during capacitor demonstration

Name of the DP			Bag DT, Tigaon Beed		
Name of the farmer			Bhagvat Vishwanath Shendge		
Pump HP			3 HP		
Voltage Measurement					
Before Capacitor installation			After Capacitor Installation		
Vry	Vyb	Vby	Vry	Vyb	Vby
418	416	419	429	425	421
Current Measurement					
Before Capacitor installation			After Capacitor Installation		
Ir	Iy	Ib	Ir	Iy	Ib
9.8	10.1	10	8.7	9.1	8.5
Power Factor Measurement					
Before Capacitor installation			After Capacitor Installation		
0.87			0.96		



Figure 3.7 Capacitor readings taken along with farmers

3.2 DTs selected for Components F

Several DTs were selected for structured demonstration of the value of capacitors. The details of selected DTs for both Beed and Wardha districts are as follows:

Table 3.3. Details of selected DTs for Beed district

Sr. No.	Village	Name of the DT	kVA rating	No. of Pumps	Common pump HPs	Problem Associated
1	Bagpimpalgaon	Canal DT	100 kVA	25	3, 5, 7.5	Overloading
		Umapur fata DT	100 kVA	20	3, 5	Low voltage
2	Rui	Baman mala DT	63 kVA	21	3, 5	Overloading
3	Tigaon	Bag DT	100 kVA	27	3, 5, 5/8, 6/12	Low Voltage
4	Shivni	Kakarwadi DT	63 kVA	19	3, 5	Low voltage
5	Pimpalwandi	Talav DT	100 kVA	16	5, 7.5, 10	Low voltage
6	Ekruka	Vashisht Kedar DT	100 kVA	25	3, 5, 6/12	Overloading

Table 3.4. Details of selected DTs for Wardha district

Sr. No.	Village	Name of the DT	kVA rating	No. of Pumps	Common pump HPs	Problem Associated
1	Kanholi	Singhan DT	100	22	3, 5, 7.5	Low voltage
2	Rohana	Raut DT	100	17	3, 5	Low voltage
3	Nandpur	Dondal DT	100	18	3, 5	Low voltage
4	Junona	Giri DT	100	24	3, 5	Low voltage and long LT line

3.3 Effect of capacitor usage on irrigation Pumps

Capacitors were installed on DTs mentioned on Table 4.1 and 4.2. All the farmers on the selected DTs installed capacitors on their pumps. After installation of capacitors many farmers shared their experience of benefits due to capacitors. Few of the capacitor benefits experience shared by farmers are as follows:

- Name of the farmer : Chandrashekhar Chaudhari , Giri DT, Junona.
“ I am the last person of the Giri dp. That’s why I had an issue regarding low voltage. I got a 5 Hp authentic connection but because of low voltage, I couldn’t use the 5 Hp pump so I replaced it with a 3 Hp pump. After installing capacitors, my low voltage problem was solved. Now even the 5HP pump is running in good condition now.”
- Name of the farmers: Ashish Pokale, Raut DT, Rohana.
“DO fuse and kitkat on our DT were always used to cut before installing the capacitor. But after installing the capacitors DO and kitkat fuse does not burns. And I have observed that water flow got increased on my pump.”
- Name of the farmers: Sakharam Mugutrao, Canal DT, Bagpimpalgaon.
“Farmers on our DP had a high pump burn rate. By installing capacitors our pump burn rate is now much less. And nowadays the fuse on DT does not burn .”